BandLink firmware architecture

Layered view of the STM32U5A9 firmware, from the app screens down to the silicon. Bare-metal: no RTOS. Interrupts only set event flags; one main loop runs each module to completion, then sleeps in Stop 2 until the next interrupt. Each layer only calls the layer below it.

S1 Stretch 1 S2 Stretch 2 Untagged = MVP

Application

What the wearer sees and does

Finder screen
Arrow + distance to the selected friend; switches to UWB distance when in range

Group screen
Paired friends, last-seen age, battery, signal

Pairing flow
Crown long-press → tap → confirm → key saved

Watch face + settings
Time from GPS, brightness, mode, unpair

Mode controller
Standby / Typical / Finding: sets TX interval, screen, IMU rate, UWB on/off

UI framework

LVGL 9 (MIT, C) · used only by the Application layer

Widgets + screens
`lv_arc` ring, `lv_label` distance, rotated `lv_image` arrow, `lv_list` group

Styles + theme
Dark theme for AMOLED power, round 466×466 layout, fonts in flash

Input devices S1
PEC06 → `LV_INDEV_TYPE_ENCODER` + focus group; touch → pointer indev

Timers + animation
`lv_timer_handler()` in ui_process(); `lv_tick_inc(1)` from LPTIM1

Rendering
Two partial buffers (466×46 RGB565 ≈ 43 KB each) in SRAM; NeoChrom (NemaGFX) draw unit S1, DMA2D / software fallback

Display port
`flush_cb` starts OCTOSPI DMA on TE; DMA-complete ISR calls `lv_display_flush_ready()`; round dirty areas to even x/y (CO5300 alignment, confirm)

Services

Hardware-independent logic

Link layer
Packet codec, GPS-timed TDMA slots, dedupe, relay queue S2

Crypto
AES-128-CCM encrypt + MIC, nonce = ID ‖ seq; uses HW AES

Navigation
Peer table, haversine distance, initial bearing, heading from quaternion

Pairing service
Group key + device ID exchange over NFC, writes key store

Ranging service S2
DS-TWR at 10 Hz when a peer is within ~30 m

Battery service
SOC, charge state, low-battery mode

Haptics service
Named patterns: paired, friend found, low battery

Scheduler

Bare-metal main loop and power

Main loop
Checks event flags in fixed priority order, calls each module's `*_process()` to completion

Event flags
One bit per source in a `volatile uint32_t`; ISRs set, loop clears (atomic)

Soft timers
LPTIM1 1 ms tick in Stop 2; timer list drives LVGL tick, TX slots, polling

Sleep
No flags pending → WFI into Stop 2; any EXTI/LPTIM/UART wake resumes the loop

Power manager
Load interlocks (no NFC field during LoRa TX / UWB), rail and peripheral gating

Persistent store
Wear-levelled flash: keys, IMU calibration, settings

Device drivers

One per chip

SX1262
Semtech `sx126x` driver + BUSY wait, DIO1 IRQ

MAX-M10S
UBX parser + config (rate, constellations, power-save); PPS capture

PN7150
NCI host stack (NXP MCU examples), P2P / card emulation

BNO085
CEVA sh2 library over SHTP; rotation vector, motion reports

CO5300 AMOLED S1
QSPI command set, TE-synced flush, brightness

DW3000 S2
Qorvo DW3000 API, deep-sleep / wake

DRV2605L · MAX17048 · BQ25180 S1
Small register drivers on I2C2

PEC06 crown S1
Timer encoder count + debounced push

HAL / BSP

STM32CubeU5

Buses
SPI1–3, I2C1–2, USART1–2, OCTOSPI1

DMA
GPDMA for SPI, QSPI frame flush, UART RX

Timers + EXTI
TIM2 PPS capture, TIM3 encoder, LPTIM tick, 9 EXTI lines

Security blocks
AES, RNG, PKA, hash

Graphics S1
NeoChrom GPU2D, DMA2D

System
Clocks, PWR/SMPS, RTC, flash, IWDG

Hardware

Per bracelet

STM32U5A9
Cortex-M33, 2.5 MB SRAM

Radios
SX1262 · PN7150 · DWM3000 S2

Sensors
MAX-M10S GNSS · BNO085 IMU

Interface
AMOLED · PEC06 · DRV2605L + LRA S1

Power
BQ25180 · TPS63802 · MAX17048 · 500 mAh LiPo

CROSS-CUTTING

Security
Group key never leaves the device after pairing; RDP level 1 on release builds; RNG-seeded nonces

Power policy

Every service declares its wake needs; power manager picks the deepest allowed stop mode

Logging + console

Ring-buffer log, UART console on the pogo dock, SEGGER RTT over SWD in dev

Firmware update

ST system bootloader over USART1 on the dock; BNO085 updated through BOOTN

Testing

Services build on the host (no HAL) for unit tests: codec, crypto, nav math

Rule: Application and Services never include a HAL header. Hardware access goes through a driver interface, so the same service code runs on the Nucleo, the custom PCB, and a PC test build.

Main loop modules

The loop checks flags top to bottom and restarts from the top after each module runs, so radio work is never stuck behind a UI redraw for more than one module's budget.

ORDER	MODULE	EVENT FLAG SET BY	OWNS	MAX RUN PER CALL
1	`radio_process()`	DIO1 EXTI, TDMA slot timer	SX1262 driver, link layer, crypto	<2 ms
2	`ranging_process()` S2	UWB IRQ, 100 ms timer in Finding mode	DW3000 driver, ranging service	<2 ms
3	`gnss_process()`	USART2 DMA idle-line, PPS capture	UBX parser, navigation, peer table	<1 ms
4	`imu_process()`	BNO085 H_INTN	sh2 library, heading, wrist-raise	<1 ms
5	`nfc_process()`	Crown long-press, PN7150 IRQ	NCI state machine, pairing service	<3 ms
6	`ui_process()`	LVGL timer, display TE, crown events	lv_timer_handler(), display driver, haptics	<5 ms (DMA does the flush)
7	`system_process()`	1 s timer, ALRT, charger INT	Battery service, power manager, watchdog kick, log flush	<1 ms

Rules that make bare-metal work here

- ISRs do the minimum: set a flag, grab a timestamp, start DMA. No SPI/I2C transactions inside an ISR.
- No blocking waits. Every multi-step exchange (NCI, UBX config, DW3000 ranging) is a state machine that returns and resumes on its next flag.
- Timing-critical radio actions happen in hardware: LoRa TX starts from the slot-timer ISR writing one SetTx command, and UWB replies use the DW3000's delayed-TX.
- Measure each module's worst case with a GPIO toggle on the scope; if one exceeds its budget, split it into steps.

One packet's path: a friend's position reaches the screen

1	**SX1262**	Packet ends; DIO1 fires the EXTI interrupt.
2	**ISR → loop**	ISR sets EVT_RADIO and returns; the core wakes from Stop 2, the main loop sees the flag first and calls radio_process(), which reads the FIFO over SPI1.
3	**link + crypto**	Checks seq against the dedupe table, verifies the AES-CCM MIC with the group key, decrypts. Bad MIC → dropped.
4	**link (S2)**	If relay is on and the hop count allows, queues a re-broadcast for the next free slot.
5	**navigation**	Updates the peer table; computes distance and bearing from this bracelet's last GPS fix.
6	**sensor**	Latest BNO085 heading gives the arrow angle: bearing − heading.
7	**ui**	Navigation sets EVT_UI; ui_process() invalidates the arrow and distance widgets, and LVGL redraws only those areas on the next TE pulse.
8	**haptics**	If this friend just came back into range, plays the "found" pattern.